

LAB REPORT

Lab Title	Interactive Input and Selections	Lab No.	2
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Lab description/objectives:

1. Demonstrate understanding of interactive input handling, selection statement (if-else, switch) and logical operators in C programming.
2. Implement percentage-to-grade conversion and conditional message printing based on character input as defined in the lab tasks/requirements.
3. Analyze input validation, handling of invalid entries, and proper use of conditional logic to ensure program robustness.

Source code:**Task 1:**

```
#include <stdio.h>
int main(){
    double i;
    int temp = scanf("%lf", &i);
    if (!temp){
        printf("Invalid Value");
        return 0;
    }
    if (i<60){
        printf("Your grade is F.");
    }else if (i<70){
        printf("Your grade is D.");
    }else if (i<80){
        printf("Your grade is C.");
    }else if (i<90){
        printf("Your grade is B.");
    }else if (i<=100){
        printf("Your grade is A.");
    }else {
        printf("Invalid value.");
    }
    return 0;
}
```

Task 2:

```
#include <stdio.h>
#include <ctype.h>
```

```
int main(){
    char score;
    scanf("%c", &score);
    score = toupper(score);
    switch (score)
    {
    case 'A':
    case 'B':
        printf("Excellent!");
        break;
    case 'C':
    case 'D':
        printf("Good");
        break;
    case 'E':
    case 'F':
        printf("Need Improvement");
        break;
    case 'O':
        printf("Outstanding!");
        break;
    case 'U':
        printf("Undefined grade");
        break;
    default:
        printf("Invalid grade");
        break;
    }
    return 0;
}
```

Program output:

Task 1:

```
78
Your grade is C.
```

```
r
Invalid Value
```

```
180
Invalid value.
```

Task 2:

```
a  
Excellent!
```

```
0  
Outstanding!
```

```
10  
Invalid grade
```

Analysis:

1. Validate Range of Inputs

- The range of task 1 is 0-100. If we input the number out the range or input other character, the input is invalid.
- The range of task 2 is those letters in either uppercase or lowercase. If we input others then the input is invalid.

2. Optimize

- In task 1 we choose the right scope to decrease the number of judgements.
- In task 2 we do not use “break” in some cases and use the function to transfer the lowercase letter to uppercase letter to get our wanted result.

3. Manage Edge Cases:

- In task 1, we check the return value of the function “scanf()” to ensure the right input. Then we eliminate those invalid cases.
- In task 2, we set default case as the invalid input to ensure the right input.

Conclusion:

- When doing the interactive input, we need to ensure the right format of the input and cope with the input in advance.
- When doing selection, we need to set the appropriate scope and optimize the cases so that we can get the right output more accurately and rapidly.